CSE4078S26_Grp6 Progress Summary

Spring 2026 Term Project - Turkish Small LLM Baseline Evaluation and Fine-Tuning Plan

Group	Group 6
Course	CSE4078 - Spring 2026 Term Project
Project Task	Evaluate two small open-source LLMs for Turkish and select one model for supervised fine-tuning.
Group Members	To be filled with member names before submission

1. Selected Baseline Models

Two 3B-class instruction-tuned open-source models were selected for the baseline evaluation. Llama-3.2-3B-Instruct was initially considered, but access to its gated Hugging Face repository was still pending. Therefore, amd/Instella-3B-Instruct was used as an accessible 3B instruction-tuned alternative for the progress deadline.

Model ID	Type	Purpose
Qwen/Qwen2.5-3B-Instruct	3B-class instruction-tuned causal LM	Main baseline and selected fine-tuning candidate
amd/Instella-3B-Instruct	3B-class instruction-tuned causal LM	Second baseline comparison model

2. Dataset Usage

Because Group 6 is an even-numbered group, the mandatory dataset is Renicames/turkish-law-chatbot. The dataset contains approximately 14.9k Turkish legal question-answer examples, with a 13.4k training split and a 1.5k test split. The test split is reserved for baseline and final evaluation, while the training split is used as the SFT corpus for fine-tuning.

Item	Description
Test corpus	Renicames/turkish-law-chatbot test split. Used for baseline evaluation. The progress evaluation used 200 test samples.
SFT corpus	Renicames/turkish-law-chatbot train split. Used for supervised fine-tuning. The test split will remain unseen during training.
Task format	Turkish legal question answering. Each sample contains a user question and a reference legal answer.

3. Baseline Evaluation Setup

Both baseline models were evaluated using the same test subset and the same generation settings to ensure a fair comparison. Greedy decoding was used to make the outputs deterministic. The generated answers were compared against reference answers using ROUGE and BERTScore.

Setting	Value
Evaluation samples	200 samples from the test split
Generation setting	Greedy decoding, do_sample=False
Maximum generated tokens	128 max_new_tokens
Memory optimization	4-bit quantization
Metrics	ROUGE-1, ROUGE-2, ROUGE-L, ROUGE-Lsum, and BERTScore Precision/Recall/F1

4. Baseline Results

The table below summarizes the preliminary baseline results on 200 test samples. Qwen/Qwen2.5-3B-Instruct achieved substantially higher ROUGE and BERTScore values than amd/Instella-3B-Instruct on the tested subset.

Model	Samples	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	BERTScore F1
Qwen/Qwen2.5-3B-Instruct	200	0.1802	0.0766	0.1463	0.1468	0.6347
amd/Instella-3B-Instruct	200	0.0224	0.0001	0.0201	0.0202	0.5231

Additional BERTScore Details

Model	Precision	Recall	F1	BERTScore Model
Qwen/Qwen2.5-3B-Instruct	0.6151	0.6583	0.6347	bert-base-multilingual-cased
amd/Instella-3B-Instruct	0.5083	0.5426	0.5231	bert-base-multilingual-cased

5. Selected Model for Fine-Tuning

Based on the baseline results, Qwen/Qwen2.5-3B-Instruct is selected for fine-tuning. It achieved higher scores across all reported ROUGE metrics and BERTScore F1 on the 200-sample baseline evaluation subset.

6. Fine-Tuning Plan

The selected model will be fine-tuned using the Renicames/turkish-law-chatbot training split as the supervised fine-tuning corpus. The test split will remain unseen until final evaluation. A parameter-efficient method, QLoRA, will be used to reduce GPU memory requirements. For training-time validation, a small validation subset will be created from the training split only, preserving the official test split for unbiased evaluation.

Component	Planned Approach
Fine-tuned model	Qwen/Qwen2.5-3B-Instruct
SFT corpus	Renicames/turkish-law-chatbot training split
Method	QLoRA / LoRA-based supervised fine-tuning
Validation	Validation split created from training data only
Final evaluation	Re-evaluate the fine-tuned model on the same public test split and compare before/after results

7. Current Repository Status

The repository contains scripts for dataset preparation, inference, ROUGE evaluation, BERTScore evaluation, and QLoRA fine-tuning. The next phase will focus on full-scale evaluation, fine-tuning the selected model, final before/after comparison, and error analysis.

Note

The reported numbers are preliminary progress-deadline results based on 200 test samples. Full 1,500-sample evaluation and fine-tuning results will be reported in the final submission.